# Title.

*" Computer Based Test(CBT) Platform embedded with AI proctoring(AI ExamGuard)"*

# Problem Statement

Current Online examination system consist of various problems:

- No Real Time monitoring
- Manual Invigilation dependency is high
- High unfair means chances
- Less Scalable for huge set of students
- High network usage due to live video transmission.

That's why there is a need for intelligent , secure and scalable platform that can ensure exam is conducted properly and efficiently.

# Introduction

This project is an online examination platform specifically designed for conducting CBT (computer based test ) examination . Current online examination platform lacks monitoring, highly costly to conduct and manage which makes them prone to use of unfair means through phones ,books or tabs switching.Project is divided in three parts **Frontend (UI) , Ai proctoring Layer ,Backend** .This entire project based on **MERN** stack ( MongoDB ,Express.js ,React.js ,Node.js). since its open source ,easy to learn and cost effective hence it is the most loved tech stack by the developers to build an entire full-stack project. As we are using express.js the project is scalable and able to handle multiple users request. Proper **authentication** and **authorization** is handled by JWT( Jason web token) and apart from this **Tensorflow.js , MediaPipe** are used for image detection and recognition which continuously monitors user and its behaviour through inbuilt camera or webcam.

# Objective

- Provide a platform for conducting online examination.
- Easy to use and user friendly responsive UI.
- Secure back-end for exam integrity.
- Store exam data ,result data efficiently for fast retrieval.
- Role based frontend i.e  user or examiner.
- Scale for large no of students.

# Scope of Project

- Student /Examiner registration and authentication.
- Timely and Efficient Exam Conduction.
- Automated evaluation tests.
- Real Time Ai Monitoring.
- Suspicious Activity Logging.
- Examiner Dashboard for managing Tests and examinee.

# Project Architecture

Project is based on MVC ( Model-Views-Controller) architecture .It is the most used industry standard framework [1] .It separates ui/ux (front-end) part into **Views** , business logic (back-end) and database schema (back-end)  into **Model** and **Controller** acts as a bridge between model and views.

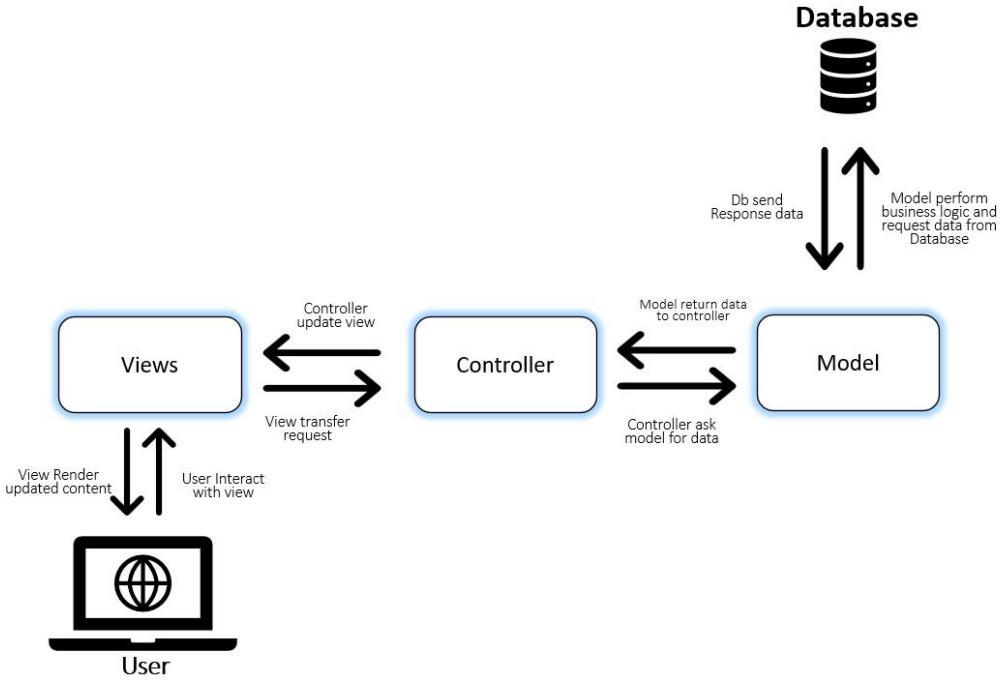
Exam Interface and monitoring through AI is handled in frontend.

AI model locally process the data in browser through webcam.

If any suspicious activity are generated those are stored into database.

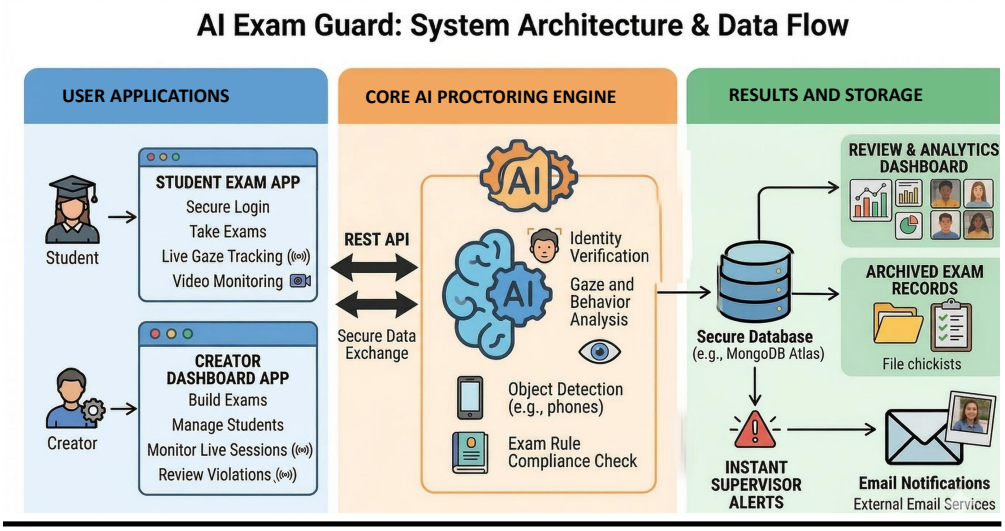
All exam Result are stored though back-end into MongoDB

- This makes managing and updating code efficient.
- Tracking bugs easier .
- Increases cohesion and reduces coupling.




MVC Architecture Diagram

# Project Flow



# Module Breakdown

•**Authentication Module** — JWT-based login and registration for both creator and student roles. Role-scoped tokens stored under separate localStorage keys enable simultaneous sessions.

•**Test Creator Module** — Multi-section test builder with per-question MCQ and numerical types, marks, negative marking, section timers, and strict mode configuration. Full CRUD with publish/unpublish toggle.

•**Enrollment Module** — Students join tests using alphanumeric codes. Enrollment records link student accounts to specific tests and are used to populate the Student Dashboard.

•**Exam Engine Module** — Fullscreen-locked exam interface with real-time total and section timers, strict section navigation, question palette, answer state management with ref synchronization, and beforeunload beacon for safe submission.

•**AI Proctoring Module** — TensorFlow.js and COCO-SSD models running client-side on sampled camera frames. Violations dispatched to a shared handleDetect callback for categorization and display.

•**Scoring Module** — Server-side score calculation with positive and negative marking per question. Returns score, percentage, totalMarks, passed, and timeTaken on the Attempt document.

•**Analytics Module** — Per-test results panel with score band distribution, pass rate, average score, and per-student attempt history with violation summaries.

•**Deployment Module** — GitHub-connected CI/CD with Vercel for frontend and Railway for backend. MongoDB Atlas for managed cloud database. Environment variables for all secrets.

# Tools and Technologies

**1.React.js** – React is a front-end **library** used for designing user friendly and responsive websites. This is **component** [2] **based** library which combines functionalities of html and java-script code .Due to its component based same codes can be reused multiple times .

**2.Tailwind**- it is built on top of the CSS and is a **utility-first** [3] css framework which provides premade classes for css and can be used directly into html code without leaving html file.

**3. Node.js** – It provides **javascript runtime environment** which lets us run javascript code outside the web browser. Its Built on top of the google chromes V8 engine and lets us use javascript for creating servers and web application.

**4.Express.js** – Express is a minimal and flexible framework [4]for creating servers and web application and it is built on top of node.js which makes creating and handling more easier.

**5.MongoDb**- Since industrial data transferring format is json Mongodb lets us store data into JSON format i.e with **key value pair** for better storage and retrieval.

**6.JWT-** To enhance **authentication** and **authorization** we use jsonwebtoken which creates a **digitally signed tokens** [5] used for transmitting information between client and server.

**7.TensorFlow.js** - A Machine learning Library[6] developed by Google in 2015 and is open source to use . It lets the user build and train AI models using neural network and lets the user run theese model on browser.

**8.MediaPipe-** It is framework also developed by google and its open source ,its is used to **process real time multimedia**[7] data for example **audio ,image ,video or sensor inputs.**

**9.Face-api.js -** Its an javascript library built on tensorflow.js and it allows face recognition and analysis.

**10.COCO-SSD -** It's a pre trained model used to detect objects in images or videos.

**11. Vercel** – It is a cloud based **hosting service** lets us build and **deploy** our entire web application and make them available and scalable over the internet .Its a paid service but we can still get small storage for hosting our application on vercel

## References

[1] MVC Architecture :- https://www.geeksforgeeks.org/software-engineering/mvc-framework-introduction/

[2] React docs - https://react.dev/

[3] Tailwind- https://tailwindcss.com/

[4] Express - https://expressjs.com/

[5]JWT - https://www.jwt.io/introduction#what-is-json-web-token

[6]TensorFlow - https://www.tensorflow.org/js/tutorials

[7]MediaPipe - https://ai.google.dev/edge/mediapipe/solutions/guide